

R Series: 'dplyr' Package

In this twenty-second article in the R, Statistics and Machine Learning series, we will continue on the exploration of handling text using the 'dplyr' package. This package is a grammar for data manipulation, and provides simple and fast functions to handle data frame-like objects.



We will use R version 4.2.2 installed on Parabola GNU/Linux-libre (x86-64) for the code snippets.

```
$ R --version
R version 4.2.2 (2022-10-31) -- "Innocent and Trusting"
Copyright (C) 2022 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)
```

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You can install and load the 'dplyr' package using the following commands:

```
> install.packages("dplyr")
Installing package into '/home/shakthi/R/x86_64-pc-linux-gnu-library/4.1'
(as 'lib' is unspecified)
--- Please select a CRAN mirror for use in this session ---
...
* copying figures
* building package indices
* installing vignettes
```

```
* testing if installed package can be loaded from temporary location
* checking absolute paths in shared objects and dynamic libraries
* testing if installed package can be loaded from final location
* testing if installed package keeps a record of temporary installation path
* DONE (dplyr)
```

```
> library(dplyr)
Attaching package: 'dplyr'
```

Consider the 'Bank Marketing Data Set' available from the UCI Machine Learning Repository available at <https://archive.ics.uci.edu/ml/datasets/Bank+Marketing>. The data set is from a Portuguese banking institution and is available freely for public research use. There are four data sets available, and we will use the `read.csv()` function to import the data from a `bank.csv` file into a data frame, as shown below:

```
> bank <- read.csv(file="bank.csv", sep=";")

> bank[1:3,]
  age      job marital education default balance housing
loan  contact day
1  30 unemployed married  primary      no    1787      no
no cellular  19
2  33  services married secondary      no    4789      yes
yes cellular  11
3  35 management single  tertiary      no    1350      yes
no cellular  16
  month duration campaign pdays previous poutcome y
1  oct         79        1    -1         0 unknown no
2  may        220        1   339         4 failure no
3  apr        185        1   330         1 failure no
```

select

You can use the `select()` function to choose specific columns from the data frame. In the following example, we select the 'age', 'job' and 'description' fields from the data set: